

Task 1 -

Roles & Responsibilities - AI ML Engineer / Day to Day Activity

Roles -

An AI/ML Engineer is someone who builds and deploys systems that can “learn” from data and make decisions or predictions. Think of them as the bridge between data science and software engineering: they take models built by data scientists and make them work in real-world applications.

Key Responsibilities -

- **Data Preparation** - Collect, clean, and organize data so it's ready for training models.
- **Model Development** - Build machine learning models (classification, regression, recommendation, etc.).
- **Model Training & Testing** - Teach the model using past data, then test how well it performs.
- **Deployment** - Put the model into production (so apps/websites can use it live).
- **Optimization** - Improve speed, accuracy, and efficiency of models.
- **Collaboration** - Work with data scientists, software engineers, and business teams to align models with business goals.
- **Monitoring** - Keep track of how models perform over time and retrain them when needed.

Day - to - Day Activities -

- **Data Collection & Understanding**
 - Gather raw data from databases, APIs, or files.
 - Understand what the data represents (customer info, sales, images, etc.).

- Identify problems in the data (missing values, duplicates, errors).
- Discuss with the team what business problem the model should solve.
- **Data Cleaning & Preparation -**
 - Clean the data (remove duplicates, fix missing values).
 - Convert data into usable formats (numbers, categories, text).
 - Split data into training and testing sets.
 - Document the preprocessing steps for future reference.
- **Model Building -**
 - Choose suitable algorithms (decision trees, neural networks, regression, etc.).
 - Write Python code to build the model.
 - Train the model using the prepared data.
 - Test the model's accuracy and performance.
- **Model Evaluation & Improvement -**
 - Compare different models to see which performs best.
 - Tune hyperparameters (adjust settings to improve accuracy).
 - Add or remove features to see their impact.
 - Record results and share with the team.
- **Deployment -**
 - Package the model into an API or service.

- Deploy it on cloud platforms (AWS, Azure, GCP).
- Integrate the model into applications (web, mobile, dashboards).
- Test the deployed model with real-time data.

- **Monitoring & Maintenance -**

- Track how the model performs in real - world use.
- Detect issues like bias, errors, or performance drops.
- Retrain the model with new data if needed.
- Update documentation and logs.

- **Collaboration & Reporting -**

- Meet with business analysts, data scientists, and engineers to align goals.
- Present findings in simple reports or dashboards.
- Suggest improvements for future projects.
- Plan next steps (new features, scaling, automation).